

Bence Hervay

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Education

Computer Science BA — University of Cambridge, Trinity College

Oct 2021 – Jun 2024

Second Best Computer Science Dissertation (92.5%) in a cohort of 120 students
Full Academic Merit Scholarship awarded to only 5 students (£60k p.a.)
Strong 2:1 overall (75% average across modules)

Advanced Mathematics Class — Fazekas Mihály Grammar School

Sep 2015 – Jul 2021

National Olympiads — 1st in Mathematics (2018); **1st in Physics** (2020); consistently top 10 throughout high school
FGC Robotics World Champions (2017); **invited to the White House** in celebration of our victory against 150+ countries
Diverse results across STEM (25+ national top-3 placements in Mathematics, Computer Science, Physics, and Robotics)
A*A*A* equivalent final grades; Medallion of Excellence for outstanding performance

Artificial Intelligence Technology Training — KÜRT Academy

Jan 2019 – Jun 2019

Project-based in-depth ML course; completed as youngest participant ever (at the age of 16)

Work Experience

Quantitative Research Engineer — AGITProp

Sep 2024 – Present

Automatic detection and statistical refinement of financially interpretable factors for covariance matrix prediction
Primary ownership of numerous key modules in the pipeline (mainly mathematical components)
Key role in project planning and coordination, contributing to firm-level prioritisation decisions
Constant expansion and maintenance of the infrastructure (e.g. quantitative modules, data pipeline, testing)

Software Developer Intern — DRW

Jun 2023 – Sep 2023

Created a fully automatic monitoring and resource allocation tool interacting with 200+ employees daily
Utilised NLP techniques to filter, classify, and distribute bug reports to maximise efficiency

Quantitative Research Intern — WorldQuant

Jun 2022 – Aug 2022

Generalised index rebalance alphas by automatic inference of rebalance schedules and methodologies
Developed a framework to mass-produce low-correlation alphas using random decision forests

Senior Group Leader — Alphacademy

Aug 2021 – Jun 2023

Taught programming to 100+ students from beginner to Olympiad level
Member of the founding team; onboarded and managed the work of 20+ teachers

Projects & Research

Rupert's Problem — C++ & TypeScript

Jun 2024 – Present

Settled the central open conjecture of a 350-year-old mathematical topic — whether every convex polyhedron fits through itself
Heavily computer-assisted proof involving 10,000+ lines of C++ utilising interval arithmetic to bound floating-point inaccuracy
Novel solutions to numerous complex sub-problems; 1,000+ hours of independent research; preprint in progress

Reinforcement Learning for 2048 (Cambridge Dissertation) — C++

Sep 2023 – May 2024

Developed the world's highest-scoring algorithm and the **fastest simulation engine** for the 2048 Puzzle Game
Reaches the **2048-tile in 1.2 milliseconds** with a **99.8% success rate** and 75% of the time even the 32768-tile
Trained a custom 200M-parameter RAMnet via self-play reinforcement learning on 150M games
Combined expectimax with locally brute-forced critical phases to maximise the probability of survival

Selected Additional Projects — Python & Java & C++

Built without relying on high-level libraries to ensure deep understanding of the fundamentals

GM-level chess engine — Awarded a research scholarship by the National Research Excellence Program
3D path tracer — Won the Cambridge rendering competition organised held across the full cohort
Project Euler — Top 0.04% of all users by solving each of the first 222 problems (~250 in total)
C++ automatic differentiation framework — Used for a custom video assistant referee
Stacked MWPC for atmospheric muon tomography — Invited to the Large Hadron Collider (CMS detector)
Handwritten digit classifier DNN with a GUI; Rubik's cube solver with a GUI; Time-travel based platformer game
Solvers for Multiplayer Pac-Man, Sudoku, Nonogram, and numerous other strategic games

Skills

Python & C++ with 200,000+ LOC experience, quantitative and statistical methods, ML, advanced DSA, low-level optimisation

Cross-disciplinary STEM expertise; data-driven solutions to theoretical and practical problems from first principles

Extracurricular

Rubik's Cube speedsolving — 14-second average; 20x national top 20
Ultimate Frisbee — UK National Championship 6th (2023)
Great Court Run — 3rd (2021); 2nd (2022); 1st and Beat the Clock (2023)
Acrobatic Basketball World Cup 4th (2018) — FaceTeam Youngsters

Activities

Regular instructor at the European Summer Program on Rationality
Events Officer of Cambridge University Hungarian Society
Captain of Trinity College Ultimate Frisbee Society
Always happy to discuss interesting puzzles and brain teasers